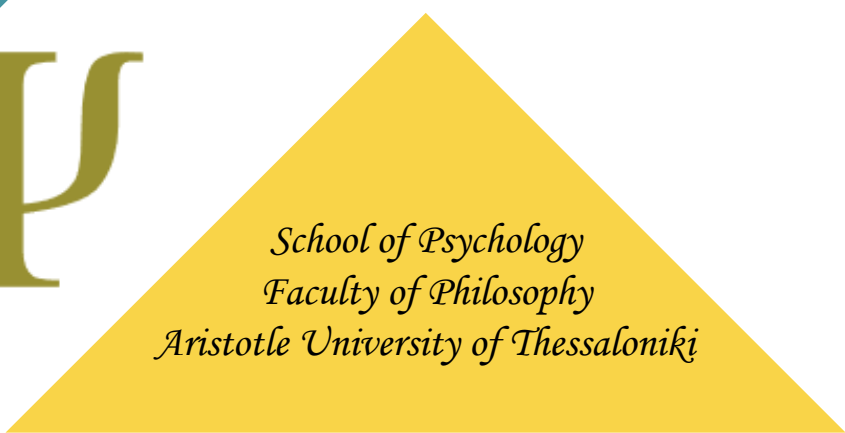




Mediation and Path Analysis

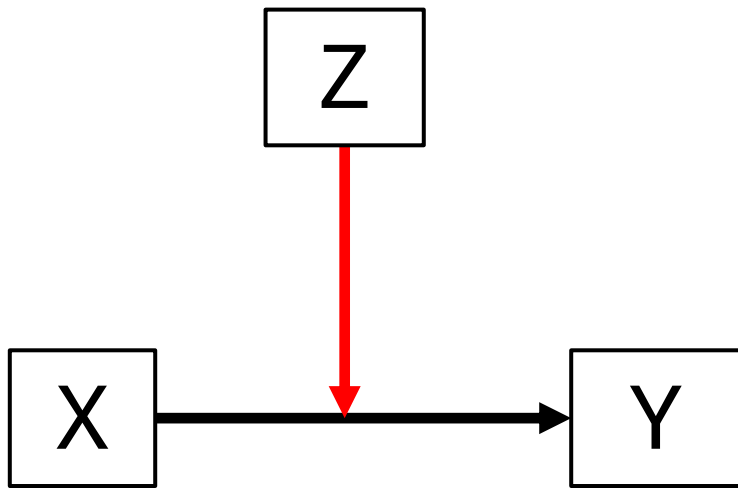
Konstantinos I. Bougioukas, MSc, PhD



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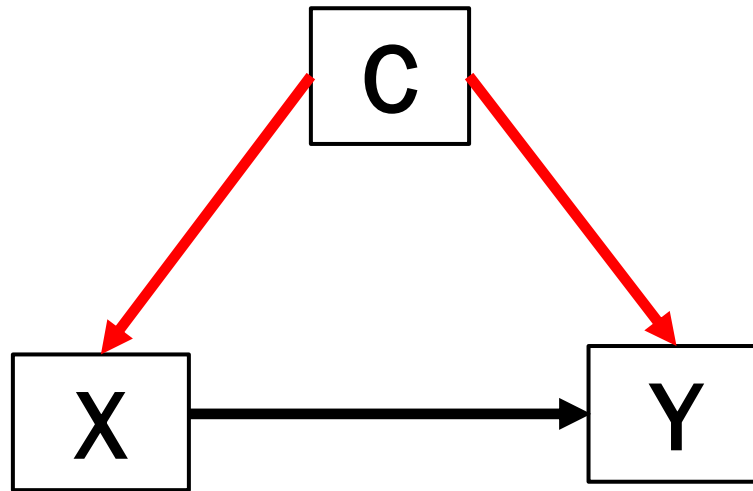
Moderator, Confounder and Mediator

Moderator



$$Y = X + Z + XZ$$

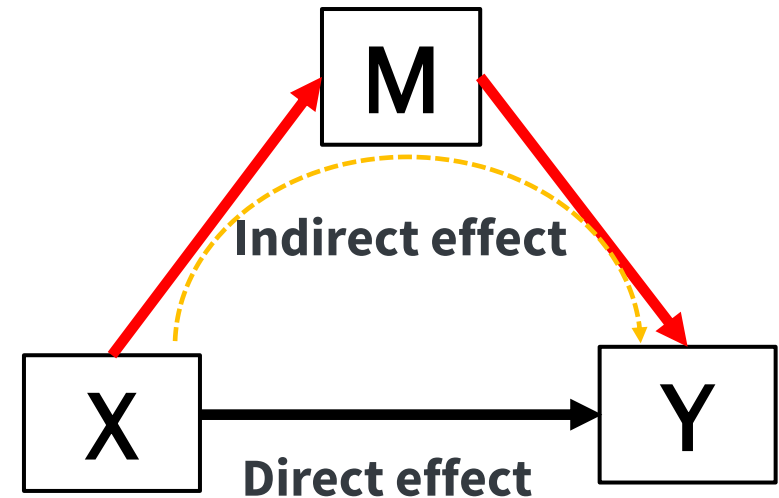
Confounder



$$Y = X + C$$

Direct effect of X on Y
after controlling for C

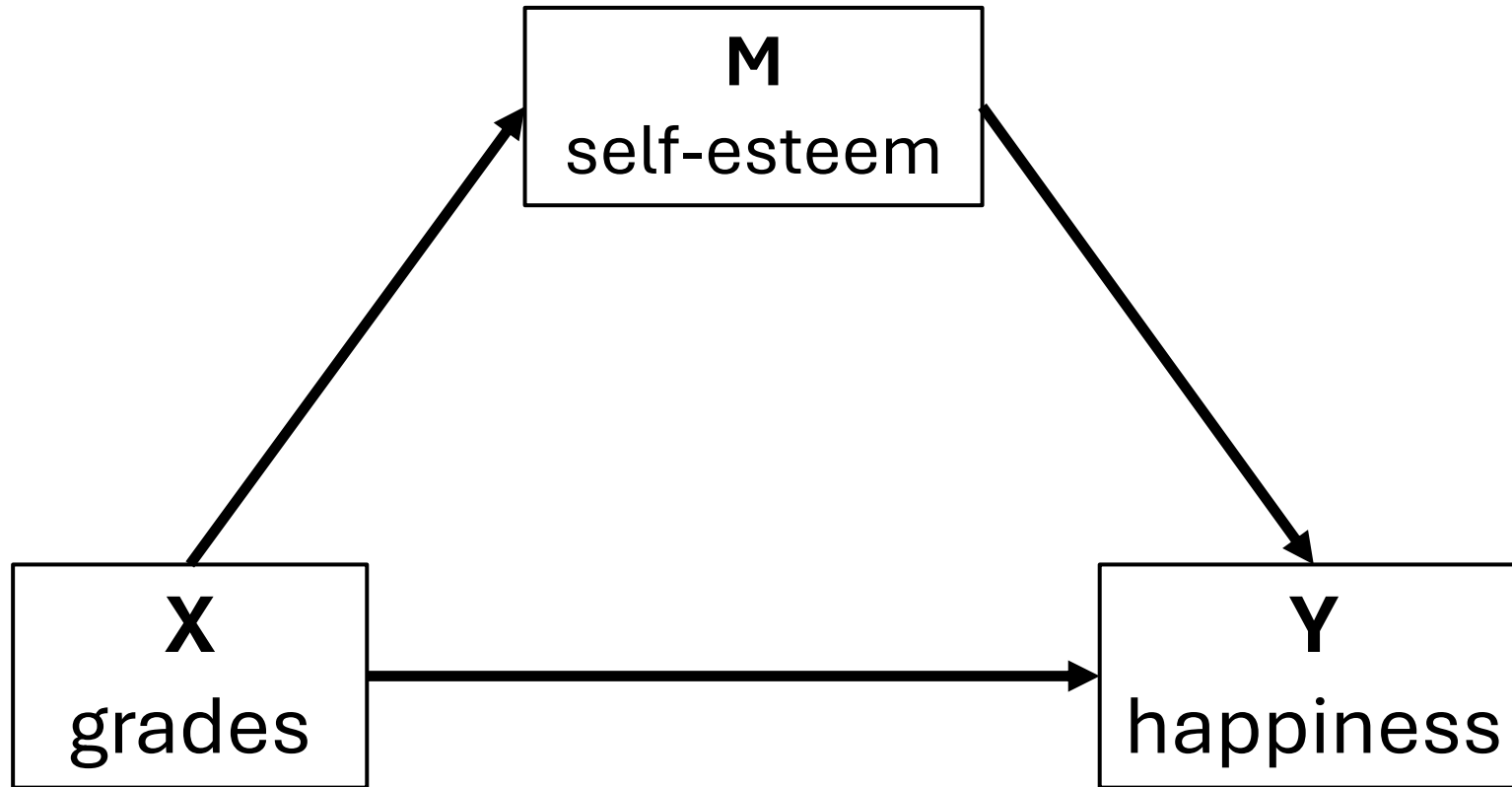
Mediator



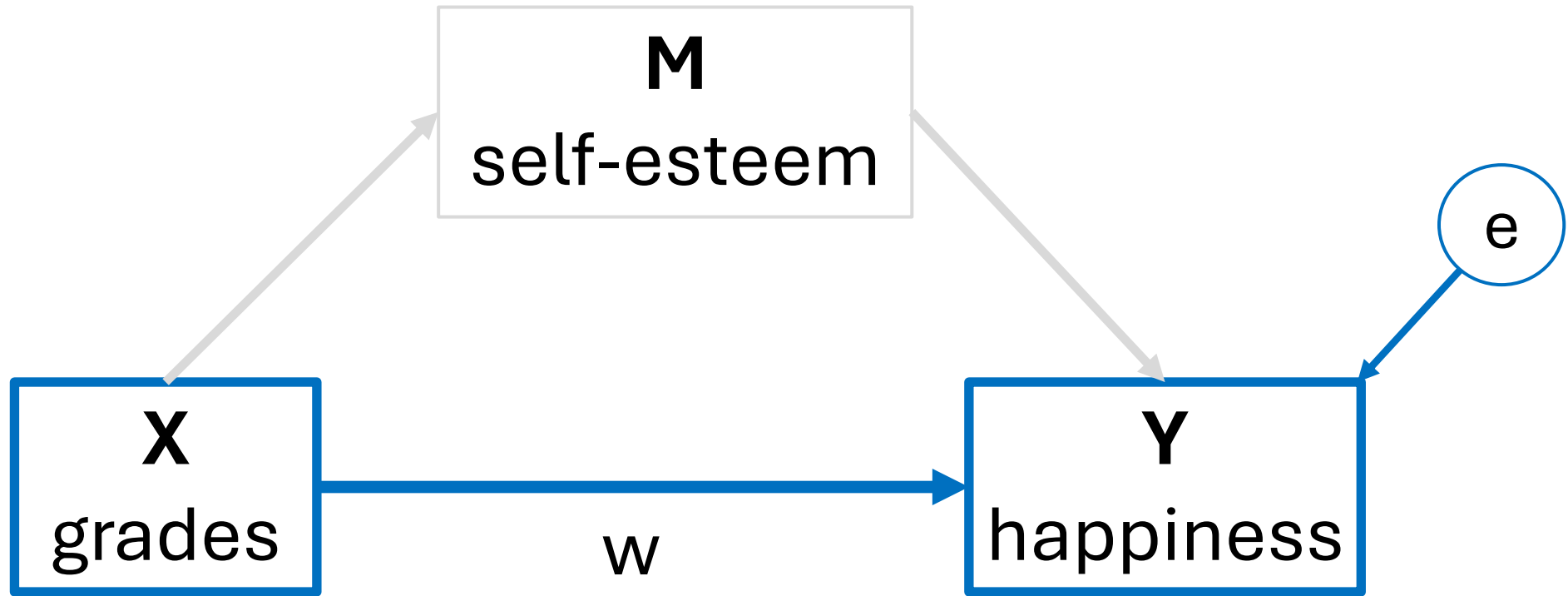
$$M = X$$

$$Y = X + M$$

A single mediation analysis

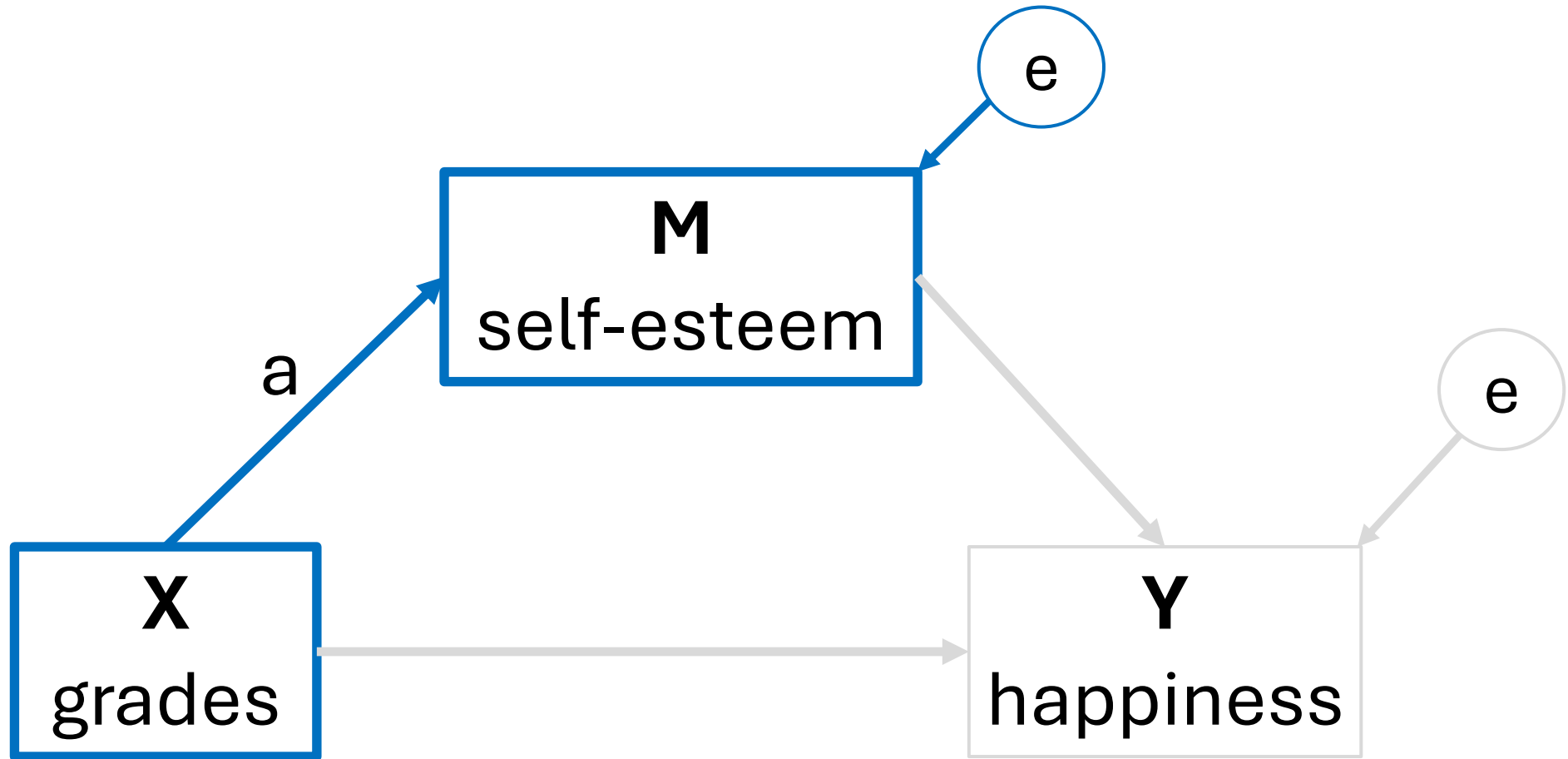


Step 1: Total effect



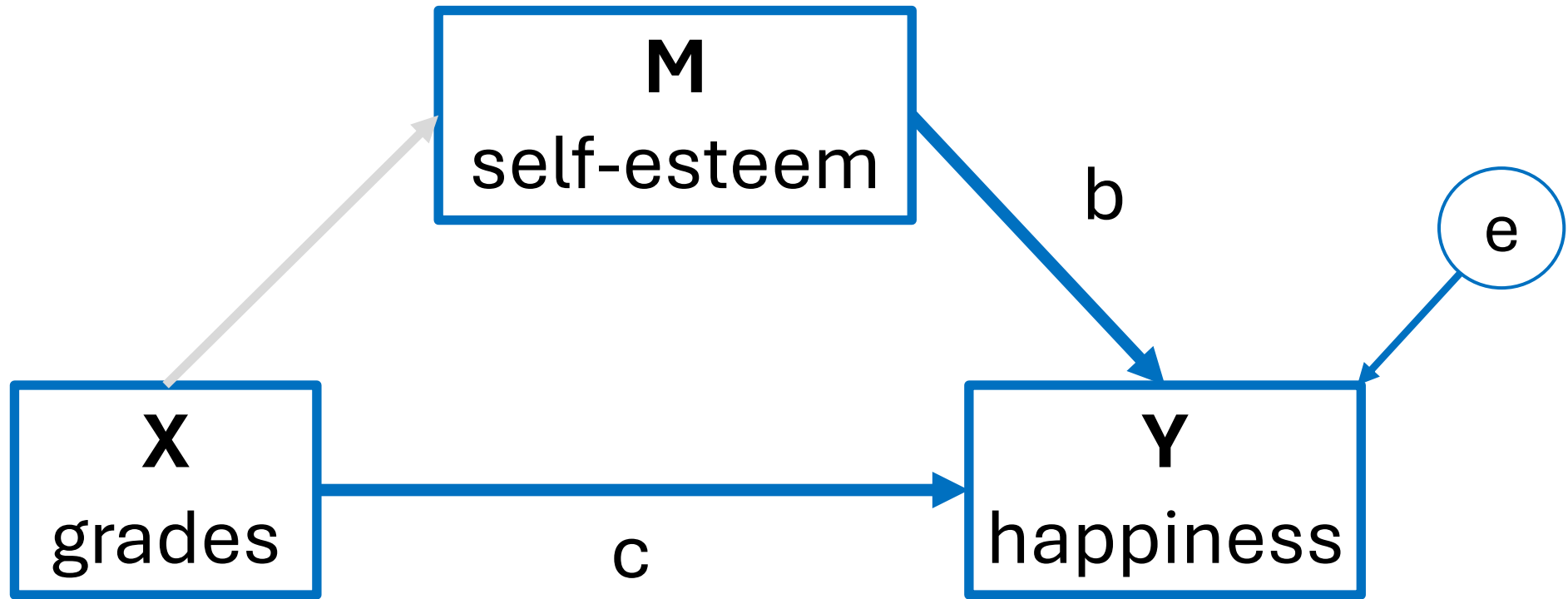
$$Y = \text{intercept} + w \cdot X + e$$

Step 2: Association between X on M



$$M = \text{intercept} + a \cdot X + e$$

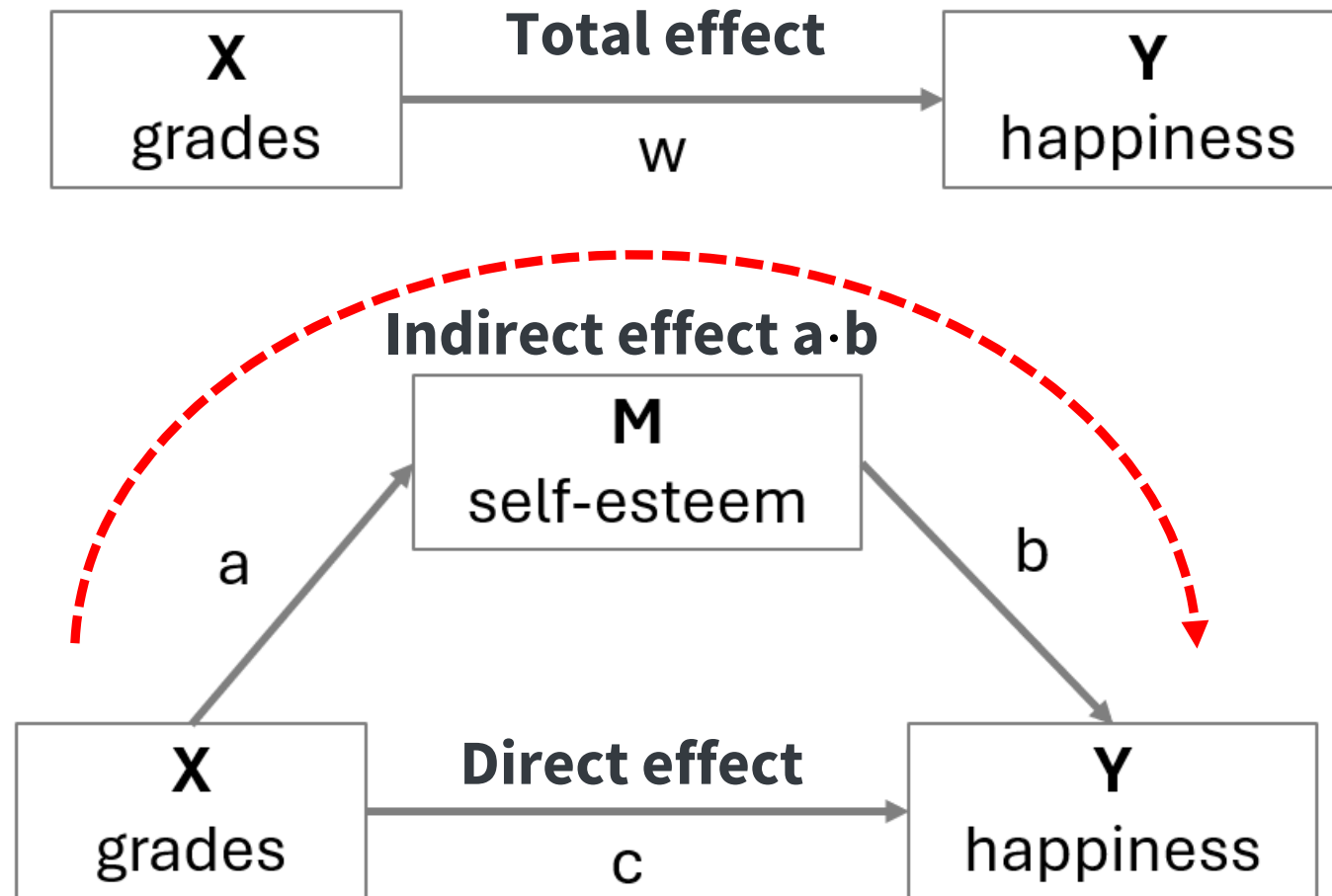
Step 3: Effects of both X and M on Y



$$Y = \text{intercept} + c \cdot X + b \cdot M + e$$

A single mediation analysis

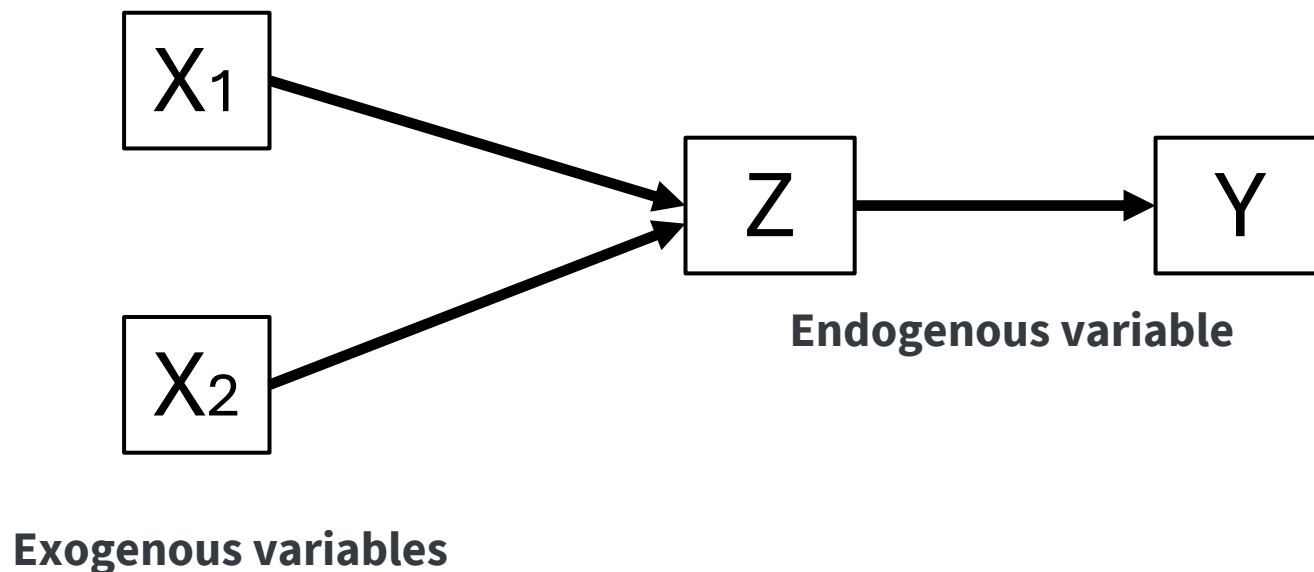
Mediation analysis decomposes the total effect $x \rightarrow y$ (**w**) into a direct effect (**c**) and an indirect effect through a **mediator** variable **M**, which is the product of **a** and **b** (**a·b**).



Path analysis

Exogenous variables: Independent variables that are influenced by factors outside the model and, in turn, influence endogenous variables. In path diagrams, **arrows originate from exogenous** variables but do not point to them.

Endogenous variables: Dependent variables that are explained by exogenous variables in the model. **Arrows point toward them** and these represent causal paths. Note that an endogenous variable may also be specified as the predictor of another endogenous variable



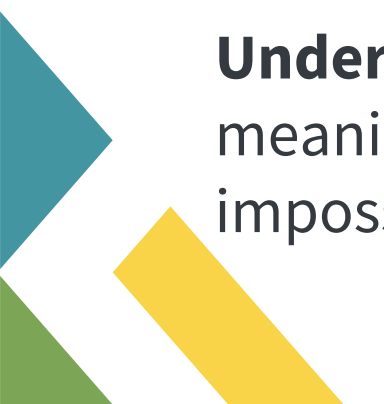
Model Identification

Model identification refers to determining whether there is sufficient information in the observed data to estimate the model's parameters (e.g., regression coefficients, variances) uniquely.

Overidentified model: when there are more unique (non-redundant) sources of information than free parameters, which means that the degrees of freedom (df) is greater than zero. This is ideal because it allows for both parameter estimation and model fit testing.

Just-identified model: it has an equal number of unique sources of information and free parameters; while the model can be estimated, its fit to the data cannot be tested.

Underidentified model: it has fewer unique sources of information than parameters, meaning it lacks enough information to estimate all free parameters, making it impossible to fit the model reliably.



The formula for calculating the number of unique (non-redundant) sources of information (**i**) available for a particular model is as follows:

$$i = \frac{p \cdot (p + 1)}{2}$$

where **p** is the number of observed variables to be modeled.

To count the number of free parameters (**k**), simply add up the number of the specified direction relations, variances, covariances, and error terms in the path analysis model.

To calculate the degrees of freedom (**df**) for the model, subtract the number of free parameters from the number unique sources of information:

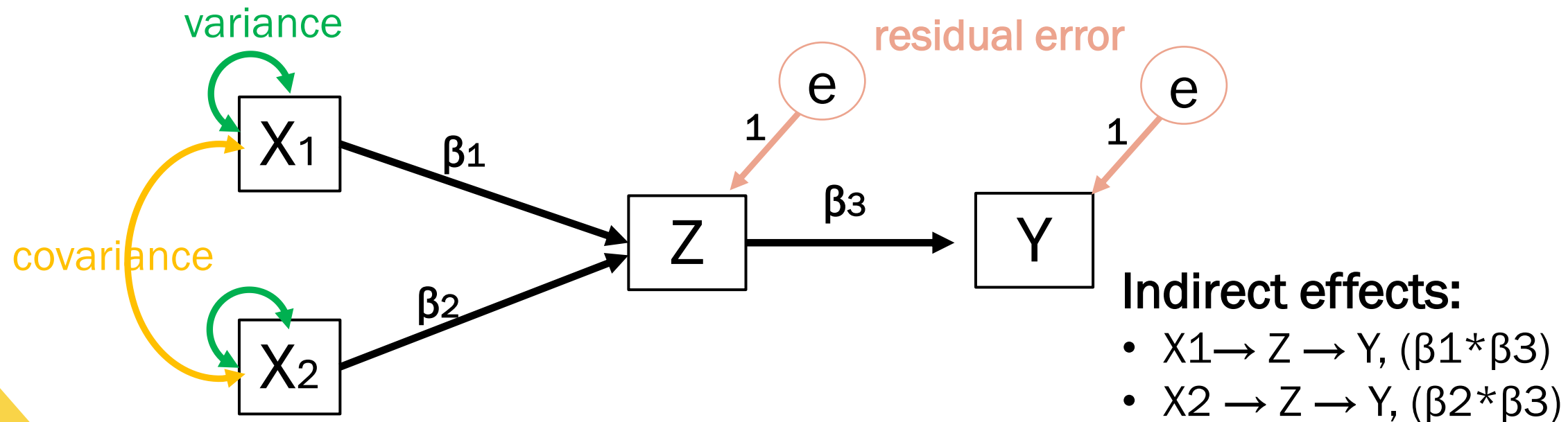
$$df = i - k$$

Path analysis (direct and indirect effects)

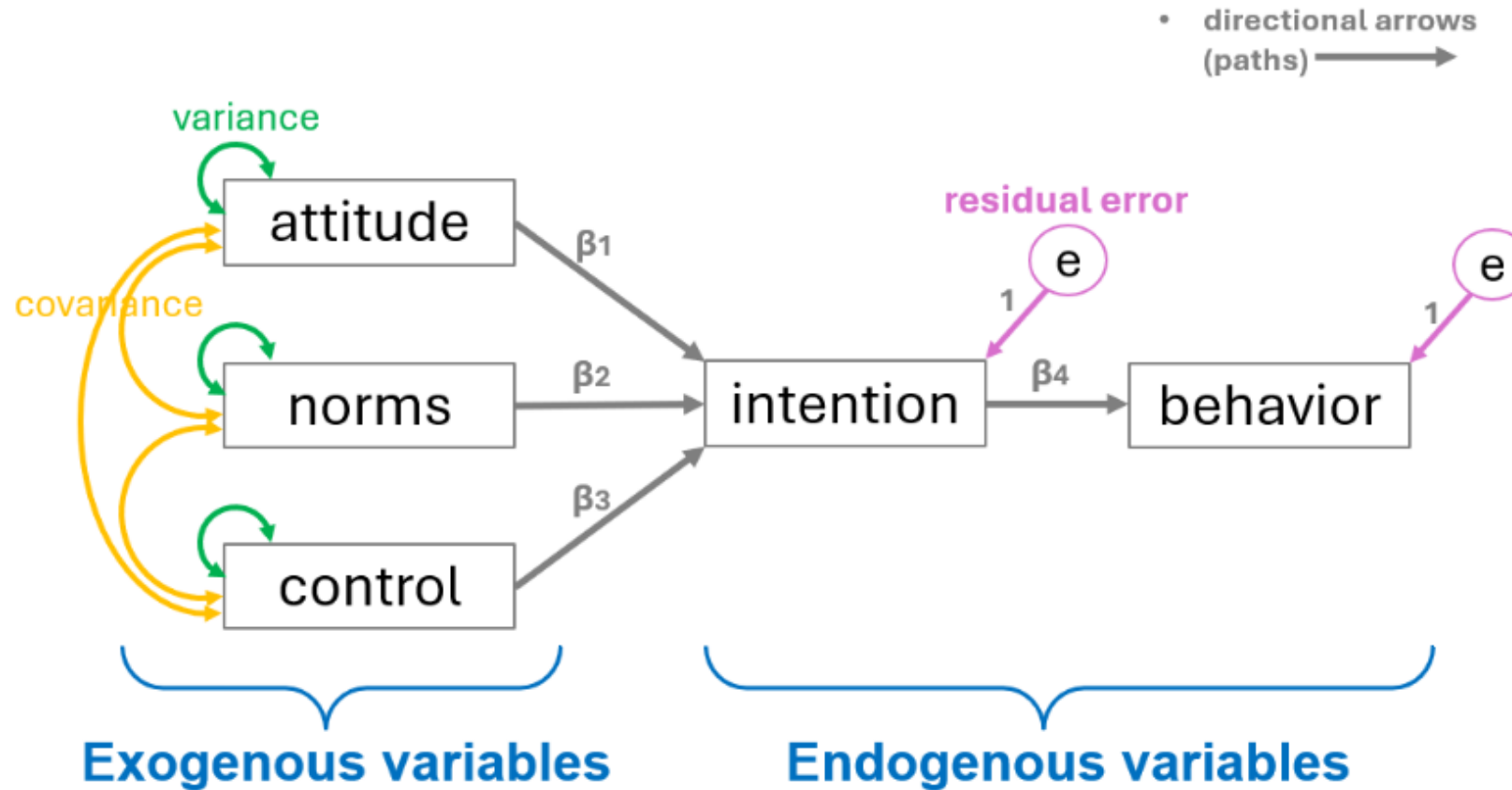
Path coefficients are regression coefficients (typically denoted as β_1 – β_3 , or sometimes as p_1 – p_3) that represent **direct effects** of one variable on another. Indirect effects of paths are calculated as the **product** of direct effects of the paths: $\beta_1 * \beta_3$ and $\beta_2 * \beta_3$.

Variances are typically represented as curved double-sided arrows, where both arrows point to the same variable. **Covariances** are also represented as double-sided arrows in which the arrows connect two distinct variables.

(Residual) error terms are added to **endogenous variables**.



Path analysis



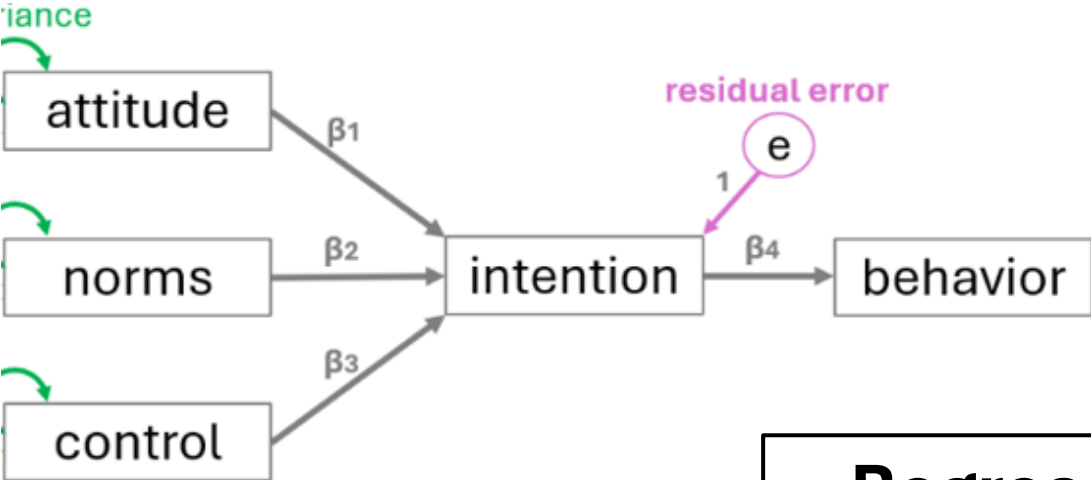
$$intention = intercept + \beta_1 attitude + \beta_2 norms + \beta_3 control + e$$

$$behavior = intercept + \beta_4 intention + e$$

Direct effects

Intercepts

	lhs	est	se	z	pvalue	ci.lower	ci.upper	std.all
intention		0.586	0.237	2.470	0.014	0.121	1.051	0.639
behavior		1.743	0.203	8.595	0.000	1.345	2.140	1.867



Regression coefficients

	lhs	rhs	label	est	se	z	pvalue	ci.lower	ci.upper	std.all
intention	attitude	b1	0.352	0.058	6.068	0.00	0.239	0.466	0.370	
intention	norms	b2	0.153	0.059	2.577	0.01	0.036	0.269	0.152	
intention	control	b3	0.275	0.058	4.740	0.00	0.161	0.389	0.291	
behavior	intention	b4	0.453	0.065	7.014	0.00	0.327	0.580	0.445	

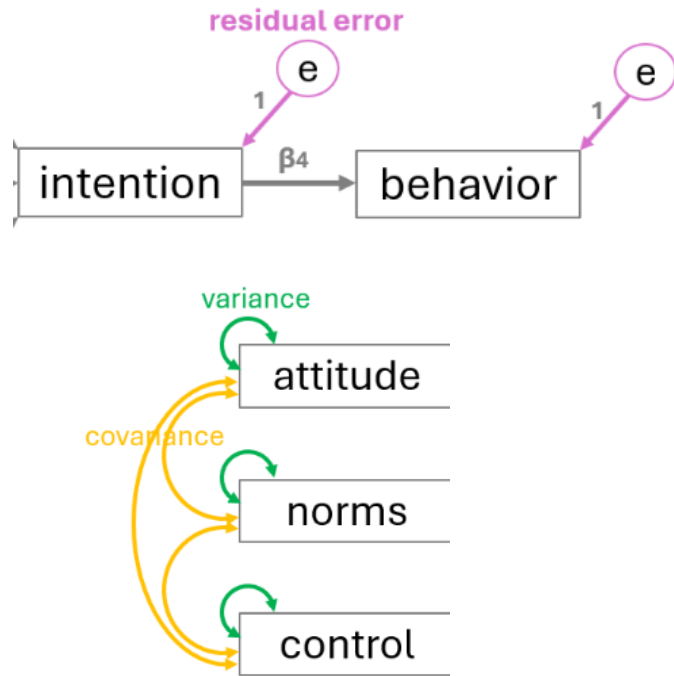
$$intention = 0.586 + 0.352 \cdot attitude + 0.153 \cdot norms + 0.275 \cdot control$$

$$behavior = 1.743 + 0.453 \cdot intention$$

Indirect effects

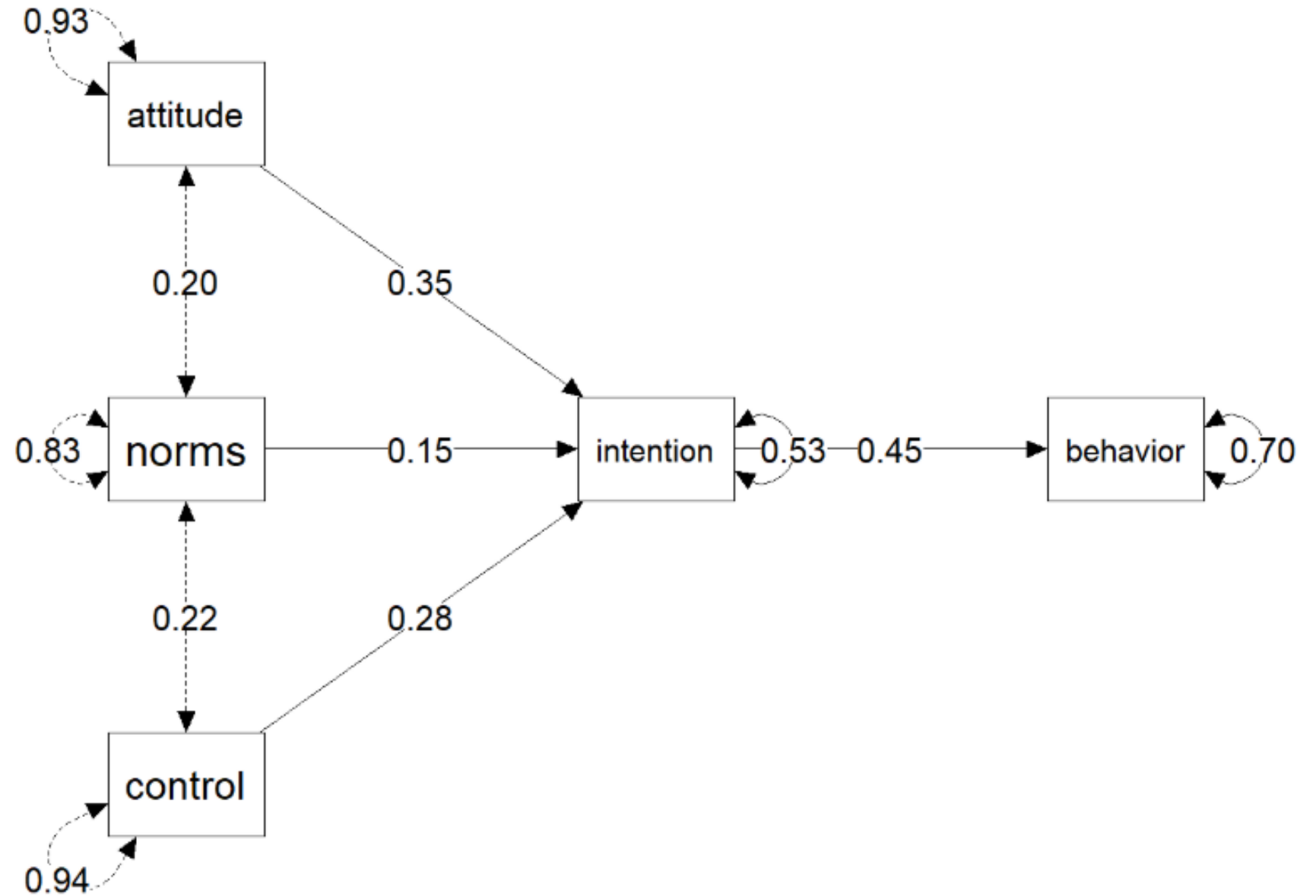
	lhs	rhs	est	se	z	pvalue	ci.lower	ci.upper	std.all
	indirect_att	b1*b4	0.160	0.035	4.589	0.000	0.092	0.228	0.165
	indirect_norm	b2*b4	0.069	0.029	2.419	0.016	0.013	0.125	0.067
	indirect_ctrl	b3*b4	0.125	0.032	3.927	0.000	0.062	0.187	0.129

Variances, covariances, and variance of errors



lhs	rhs	est	se	z	pvalue	ci.lower	ci.upper	std.all
intention	intention	0.530	0.053	9.975	0	0.426	0.634	0.631
behavior	behavior	0.699	0.070	9.975	0	0.561	0.836	0.802
attitude	attitude	0.928	0.000	NA	NA	0.928	0.928	1.000
attitude	norms	0.200	0.000	NA	NA	0.200	0.200	0.227
attitude	control	0.334	0.000	NA	NA	0.334	0.334	0.357
norms	norms	0.830	0.000	NA	NA	0.830	0.830	1.000
norms	control	0.220	0.000	NA	NA	0.220	0.220	0.249
control	control	0.939	0.000	NA	NA	0.939	0.939	1.000

Path diagram



Measure of Fit in path analysis

- Chi-square test

H_0 : The model fits the data perfectly

- Approximate Fit Index

Incremental
CFI, TLI

Absolute
RMSEA, SRMR

Criteria

Standardized Estimates
significant, strong

Non-significant chi-square

$$CFI \geq 0.95$$

$$RMSEA \leq 0.05 \text{ (but definitely } < 0.10)$$

$$SRMR \leq 0.08 \text{ (but definitely } < 0.10)$$

Combination rule:
 $CFI \geq 0.95$ and
 $SRMR \leq 0.08$